

EXHIBIT E

Curriculum Vitae

Frank Ernst

Dr. rer. nat. habil.

Leonard Case Jr. Professor of Engineering

Department of Materials Science and Engineering

Case Western Reserve University

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August 3, 2026

Academic Degrees

1997	Dr. rer. nat. habil. (Habilitation*)	Materials Science	University of Stuttgart, Germany (Advisors: E. Arzt, F. Aldinger, M. Rühle)
1987	Dr. rer. nat. (Ph. D.)	Physical Metallurgy	University of Göttingen, Germany (Advisor: P. Haasen)
1984	Dipl.-Phys. (Masters)	Physics	University of Göttingen, Germany (Advisor: P. Haasen)

Bio Data

Date of Birth	November 17, 1959
Place of Birth	Göttingen, Germany
Spouse	Sabine Ernst
Children	Meike Ernst (October 20, 1990) Oliver Ernst (September 28, 1992)

*The Habilitation is the highest academic degree in Germany. It is a post-doctoral qualification awarded for demonstrating a special qualification in research and *teaching* at the university level.

**The *venia legendi* is the permission to teach at a German university. It *can* be granted on the basis of a Habilitation.

Appointments

2016–2024	Department Chair Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
2016–2021	Faculty Director Swagelok Center for Surface Analysis of Materials CWRU, Cleveland, (Ohio), USA
Since 2012	Secondary Professor Department of Chemical and Biomolecular Engineering CWRU, Cleveland, (Ohio), USA
2008–2011	Director Case Center for Surface Engineering CWRU, Cleveland, (Ohio), USA
2008–2016	Co-Director Swagelok Center for Surface Analysis of Materials CWRU, Cleveland, (Ohio), USA
Since 2007	Leonard Case Jr. Professor Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
Since 2000	Professor Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
1997–2000	Lecturer University of Stuttgart, Germany
1989–2000	Senior Scientist Max-Planck-Institut für Metallforschung Stuttgart, Germany
1987–1988	Post-Doctoral Researcher Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA

Awards

- 2025 **Nomination for John S Diekhoff Award for Distinguished Graduate Teaching**
CWRU
- 2019 **Jessica Melton Perry Award for Disnguished Teaching in Disciplinary & Professional Writing**
CWRU
- 2019 **Graduate Teaching Award**
Case School of Engineering
- 2018 **Nomination for the Graduate Teaching Award**
Case School of Engineering
- 2017 **John S Diekhoff Award for Distinguished Graduate Teaching**
CWRU
- 2016 **Technical Educator Award**
CTSC (Cleveland Technical Societies Council)
- 2010 **Acta Materialia Top Reviewer 2009**
Elsevier
- 2010 **Materials Science and Engineering Outstanding Reviewer 2009**
Elsevier
- 2009 **Materials Science and Engineering Outstanding Reviewer 2008**
Elsevier
- 2006 **Case School of Engineering Outstanding Teacher 2005-2006**
CWRU
- 2005 **Nomination for the
2005 Carl F. Wittke Award for Excellence in Undergraduate Teaching**
CWRU
- 2005 **2004-2005 Case School of Engineering Teaching Leader**
CWRU
- 2004 **Tenure Appointment**
CWRU
- 2001 **Werner Köster Prize 2000**
German Materials Research Society (DGM) and Zeitschrift für Metallkunde
- 1997 **Venia Legendi****
University of Stuttgart
- 1993 **Tenure Appointment**
Max-Planck Society, Max-Planck-Institut für Metallforschung
- 1985 **Graduate Student Research Award**
German Materials Research Society (DGM)
- 1984 **Heineman Scholarship for Graduate Studies**
Minna-James-Heineman-Foundation (for 3 years)

Professional Activities

Professional Memberships

ASM	ASM International
MSNO	Microscopy Society of Northern Ohio
MSA	Microscopy Society of America
ACerS	American Ceramic Society
TMS	The Minerals, Metals & Materials Society
DGE	German Electron Microscopy Society
DGM	German Materials Research Society

Memberships on Boards of Scientific Journals

2010 – today	Metals: Editorial Board
2006 – 2016	International Journal of Materials Research: Advisory Board
2004 - 2013	Materials Science and Engineering A: Editorial Board
1997 – 2006	Zeitschrift für Metallkunde: Advisory Board
1996 – 2002	Journal of Electron Microscopy: Editorial Board

Service as Reviewer

For Scientific Publications

Acta Crystallographica	Acta Materialia
Advanced Materials	Applied Physics Letters
International Journal of Materials Research	Journal of Biomedical Materials Research
Journal of Electron Microscopy	Journal of Physical Chemistry
Journal of Materials Research	Materials Science and Engineering
Materials Science and Technology	Metallurgical and Materials Transactions
Metals	Microscopy & Microanalysis
Nano Letters	Philosophical Magazine
Physica Status Solidi	Physical Review B
Physical Review Letters	Science
Steel Research	Surface Science
Surface Science and Technology	Ultramicroscopy
Wiley & Sons	Zeitschrift für Metallkunde

For Research Agencies

European Science Foundation
German Science Foundation (Research Proposals, Proposals for Research Centers)
Max-Planck-Society (Proposals for International Max-Planck Research Schools)
National Science Foundation
US Department of Defense

Conference Activity

1. Organizer of "Third Symposium on Surface Hardening of Corrosion-Resistant Alloys" *Materials Science & Technology 2014*. Pittsburgh, PA, USA, October 12-16, 2014.
2. Session Chair of Symposium "Grain Boundary Engineering" *XIII International Conference on Intergranular and Interface Boundaries (iib2010)*. Mie, Japan, June 27 to July 2, 2010.
3. Co-Organizer and Session Chair of the "International Symposium on Surface Hardening of Corrosion Resistant Alloys". Cleveland, May 24 to 26, 2010.
4. Co-Organizer and Session Chair of the "International Meeting on Surface Hardening of Stainless Steel". Cleveland, October 22 and 23, 2007.
5. Co-Organizer and Session Chairs of Conference on "Progress and Challenges in Understanding Materials Interfaces." *Material Science and Technology Conference and Exhibition*. Cincinnati, October 15-19, 2006.
6. Organizer and Chair of "Interfaces: A Symposium in Honor of Manfred Rühle." *Annual Meeting of the Microscopy Society of America*. Savannah, Georgia, USA, August 01-05, 2004.
7. Chair of the Symposium "Materials Design." *First Kyoto Workshop on Computational Materials Science*. Kyoto, Japan, March 4, 2003.
8. Organizer and Chair of the *MSNO Winter Symposium on Transmission Electron Microscopy*. Cleveland, Ohio, USA, February 13, 2002.
9. Chair of a Symposium on "Data Analysis and Modeling Across length Scales." *United Engineering Foundation Meeting on Integrating Materials Science into Engineering Structures and Devices*. Lake Arrowhead, California, USA, November 11-15, 2001.
10. Chair of a Symposium on "Quantitative Analysis of Interface Structure by HRTEM." *Annual Meeting of the Microscopy Society of America 2001*. Long Beach, California, USA, August 05-09, 2001.
11. Chair of a Symposium on "Wide Band-Gap Semiconductors." *Microscopy of Semiconducting Materials XII*. Oxford, United Kingdom, March 25-29, 2001.
12. Organizer and Chair of a Symposium on "Quantitative High-Resolution Electron Microscopy." *EUREM 12*. Brno, Czech Republic, July 9-14, 2000.

13. Organizer and Chair of an *International Conference on Quantitative High-Resolution Electron Microscopy*. Berlin, Germany, March 27-29, 2000.
14. Organizer and Chair of a Symposium on "Materials Research." *29th Congress of the German Electron Microscopy Society*. Dortmund, Germany, September 5-10, 1999.
15. Organizer and Chair of a Symposium on "Advances in Quantitative and Coherent High-Resolution Electron Microscopy." *14th International Congress on Electron Microscopy*. Cancun, Mexico, August 31 - September 4, 1998.
16. Organizer and Chair of the International Conference on *Frontiers of Electron Microscopy in Materials Science*. Irsee, Germany, April 19-24, 1998.
17. Organizer and Chair of a Symposium on "Interfaces and Surfaces." *Multinational Congress of the Austrian, German, and Swiss Microscopy Societies*. Regensburg, Germany, September 8-12, 1997.
18. Chair of a Symposium on "Atomic Resolution Microscopy of Surfaces and Interfaces." *Fall Meeting of the Materials Research Society*. Boston, USA, December 2-6, 1996.
19. Chair of a Symposium on "Defects at Hetero-Interfaces." *Annual Meeting of The Materials Society*. San Francisco, USA, February 27-March 3, 1994.
20. Chair of a Symposium on "Defect-Interface Interactions." *Fall Meeting of the Materials Research Society*. Boston, USA, November 29-December 3, 1993.
21. Chair of a Symposium on "Stress Relaxation and Misfit Accommodation in Epitaxial Thin Films." *Annual Meeting of The Materials Society*. San Francisco, USA, February 17-22, 1991.
22. Chair of a Symposium on "HRTEM of Defects in Materials." *Spring Meeting of The Materials Research Society*. San Francisco, USA, April 16-20, 1990.

Publications

Academic Publications

1. F. Ernst: *Struktur von Heterogrenzflächen (Structure of Heterointerfaces)*. Habilitationsschrift (Professorial Thesis), University of Stuttgart, 1997.
2. F. Ernst: *Hochauflösende Elektronenmikroskopie der Entmischung von Al-1 at% Ag at 413 K (Decomposition Kinetics of Al-1 at% Ag at 413 K, Studied by High-Resolution Electron Microscopy)*. Dissertation (Ph. D. Thesis), University of Göttingen, 1987.
3. F. Ernst: *Rekristallisationsexperimente in $\langle 100 \rangle$ -zugverformten Kupfer-Einkristallen mit künstlicher Keimbildung (Recrystallization of $\langle 100 \rangle$ -Strained Copper Single Crystals After Artificial Nucleation)*. Diplomarbeit (Masters Thesis), University of Göttingen, 1984.

Refereed Publications[§]

1. Dernek, Ozan and **Mehdi, Redad** and Yue, Weiqi and Bachman, Jonah A. and Holt, Finley R. and Ponon, Gabriel O. and Tripathi, Pawan K. and Willard, Matthew A. and Ernst, Frank and French, Roger H.: *A Perspective on Synchrotron Data Science*. Interdisciplinary Information Sciences, vol. 31, pp. 83–88, 2025.
2. **Redad Mehdi**, Rounak Chawla, Erika I. Barcelos, Roger H. French, Frank Ernst: *2D-diffractogram Analysis: Kinematic-diffraction Simulator for Neural-network Training-data Generation*. Computational Materials Science, vol. 252, pp. 113777, 2025.
3. **Sameera Nalin Venkat**, Thomas G Ciardi, Mingjian Lu, Preston C DeLeo, Jube Augustino, Adam Goodman, Jayvic Cristian Jimenez, Anirban Mondal, Frank Ernst, Christine A Orme, Yinghui Wu, Roger H French, Laura S Bruckman: *A General Materials Data Science Framework for Quantitative 2D Analysis of Particle Growth from Image Sequences*. Integrating Materials and Manufacturing Innovation, vol. 13, pp. 71-82, 2024.
4. **Weiqi Yue**, Pawan K. Tripathi, Gabriel Ponon, Zhuldyz Ualikhankyzy, Donald W. Brown, Bjorn Clausen, Maria Strantzä, Darren C. Pagan, Matthew A. Willard, Frank Ernst, Erman Ayday, Vipin Chaudhary, Roger H. French: *Phase Identification in Synchrotron X-ray Diffraction Patterns of Ti-6Al-4V Using Computer Vision and Deep Learning*. Integrating Materials and Manufacturing Innovation, vol. 13, pp. 36-52, 2024.

[§]Author names in **boldface** indicate students or post-doctoral scholars I advised.

5. **WeiQi Yue, Mohommad Redad Mehdi**, Pawan K Tripathi Matthew A. Willard, Frank Ernst, Roger H. French: *Exploring 2d X-ray Diffraction Phase Fraction Analysis With Convolutional Neural Networks: Insights From Kinematic-diffraction Simulations*. MRS Advances, vol. 9, pp. 921-928, 2024.
6. Leveraging Deep Neural Networks for Feature Extraction from 2D XRD Diffractograms Acquired at Synchrotron Beamlines: *Pawan K. Tripathi, WeiQi Yue, Redad Mehdi, Finley Holt, Gabriel Ponon, Vipin Chaudhary, Frank Ernst, Matthew Willard, Bjorn Clausen Donald W. Brown, Daniel Savage, Roger H. French*. Los Alamos National Laboratory (LANL) Report, vol. LA-UR-23-33318, pp. 1, 2023.
7. **C. Illing, Z. Ren, F. Ernst**: *Low-Temperature Carburization: Ex Situ Activation of Austenitic Stainless Steel*. Metals, vol. 2023, pp. 335-359, 13.
8. **C. Illing, M. Bestic, F. Ernst**: *Additive Manufacturing: Corrosion Proofing by Infusion of Interstitial Solute—Exemplified for Alloy 22*. Metals, vol. 2023, pp. 127-143, 13.
9. T. Li, S.C. Chien, **Z. Ren**, W. Windl, F. Ernst, and G.S. Frankel: *Understanding the efficacy of concentrated interstitial carbon in enhancing the pitting corrosion resistance of stainless steel; response to Comment by Martin et al.*. Scripta Materialia, vol. 224, pp. 115084, 2023.
10. T. Li, S.C. Chien, **Z. Ren**, W. Windl, F. Ernst, and G.S. Frankel: *Understanding the Efficacy of Concentrated Interstitial Carbon in Enhancing the Pitting Corrosion Resistance of Stainless Steel*. Acta Materialia, vol. 221, pp. 117433-447, 2021.
11. **C. Illing, Z. Ren, A. Agaponova**, A. Heuer, F. Ernst: *Rapid Alloy Surface Engineering Through Closed-Vessel Reagent Pyrolysis*. Metals, vol. 11, pp. 1764-1782, 2021.
12. **Z. Ren**, F. Ernst: *Stress–Corrosion Cracking of AISI 316 L Stainless Steel in Seawater Environments: Effect of Surface Machining*. Metals, vol. 10, pp. 1324-1341, 2020.
13. W. Niu, **Z. Li**, F. Ernst, R.S. Lillard: *The Passivity of Low-Temperature Carburized Austenitic Stainless Steel AISI-316L in a Simulated Boiling-Water-Reactor Environment*. Journal of Nuclear Materials, vol. 537, pp. 152197, 2020.
14. **Z. Ren**, F. Ernst: *High-Temperature Phase Transformations in AISI 316 Stainless Steel Infused With Concentrated Interstitial Carbon*. Journal of Alloys and Compounds, vol. 819, pp. 153000, 2020.

15. **J.C. Dalton**, F. Ernst, A.H. Heuer: *Low-Temperature Nitridation of 2205 Duplex Stainless Steel*. Metallurgical and Materials Transactions A, vol. 51, pp. 607-617, 2020.
16. **Z. Ren**, F. Ernst: *Electronic Impact of Concentrated Interstitial Carbon on Physical Properties of AISI-316 Austenitic Stainless Steel*. Acta Materialia, vol. 2019, pp. 96-105, 173.
17. **Z. Ren**, A.H. Heuer, F. Ernst: *Ultrahigh-Strength AISI-316 Austenitic Stainless Steel Foils Through Concentrated Interstitial Carbon*. Acta Materialia, vol. 2019, pp. 231-240, 167.
18. **Z. Ren**, S.Eppell, S. Collins, F. Ernst: *Co-Cr-Mo Alloys: Improved Wear Resistance Through Low-Temperature Gas-Phase Nitro-Carburization*. Surface & Coatings Technology, vol. 378, pp. 124943-1-9, 2019.
19. **X. Zhan**, M. Daniil, X. Liu, M.A. Willard, R. Akolkar, F. Ernst: *Effect of Tungsten Alloying on Magnetic Properties of Amorphous Ni-P*. Journal of Alloys and Compounds, vol. 2019, pp. 742-749, 786.
20. **Z. Li, C. Illing**, A.H. Heuer, F. Ernst: *Low-Temperature Carburization of AL-6XN Enabled by Provisional Passivation Metals* 2018 997-1011
21. **Z. Li, X. Zhan**, T. Paramasivam, W. Niu, A.H. Heuer, S. Maloy, R. Lillard, F. Ernst: *Surface Engineering of IN-718 by Low-Temperature Carburization: Properties and Thermal Stability*. Surface Engineering (2018). 35 281-293
22. **S. Lingala**, D. Jayne, F. Ernst: *Effect of Lubricant Additives on the Tribological Behavior of Aluminum Alloy Against Steel*. International Journal of Materials Research (2018). 789-802
23. W. Niu, **Z. Li**, F. Ernst, R.S. Lillard: *Passivity of Surface-Engineered Alloys in a Simulated Boiling-Water-Reactor Environment*. Corrosion, vol. 74, pp. 635-653, 2018.
24. **X. Zhan**, P. Zhang, P.M. Voyles, X. Liu, R. Akolkar, F. Ernst: *Effect of Tungsten Alloying on Short-to-Medium-Range-Order Evolution and Crystallization Behavior of Near-Eutectic Amorphous Ni-P*. Acta Materialia, vol. 122, pp. 400-411, 2017.
25. **D. Wang**, H. Kahn, F. Ernst, A.H. Heuer: *“Colossal” Interstitial Supersaturation in Delta Ferrite in Stainless Steels: (II) Low-temperature Nitridation of the 17-7 PH Alloy*. Acta Materialia, vol. 124, pp. 237-246, 2017.
26. J. Wang, **Z. Li, D Wang**, S. Qiu, F. Ernst: *Thermal stability of low-temperature-carburized austenitic stainless steel*. Acta Materialia, vol. 128, pp. 235-240, 2017.

27. **A. Zangiabadi, J. Dalton, D. Wang, F. Ernst, A.H. Heuer:** *The Formation of Martensitic Austenite during Nitridation of Stainless Steel.* Metallurgical and Materials Transactions A, vol. 48A, pp. 8-13, 2017.
28. **F. Ernst:** *Background of SAM Atom-Fraction Profiles.* Materials Characterization, vol. 125, pp. 142-151, 2017.
29. **D. Wang, H. Kahn, F. Ernst, A.H. Heuer:** *“Colossal” Interstitial Supersaturation in Delta Ferrite in 17-7 PH Stainless Steels After Low-temperature Nitridation.* Microscopy and Microanalysis (22). 2016 2020-2023
30. **L. Yu, Z. Vashaei, F. Ernst, A. Kumar, R. Akolkar:** *Direct Electroless Deposition of Nano-scale Copper Films on Ruthenium: Role of Ethylene-diamine Additive as Nucleation Promoter.* Journal of the Electrochemical Society, vol. 163, pp. D374-D378, 2015.
31. **X. Zhan and F. Ernst:** *Crystallization Micro-Mechanism of Near-Eutectic Amorphous Ni-P.* Acta Materialia, vol. 104, pp. 274-282, 2016.
32. **H. Kahn, F. Ernst, A.H. Heuer:** *Increasing the Coefficient of Sliding Friction of NiCr at Low Loads by Interstitial Surface Hardening.* Wear, vol. 346-347, pp. 1-5, 2016.
33. **D. Wu, Y. Ge, H. Kahn, A.H. Heuer, F. Ernst:** *Diffusion Profiles after Nitrocarburizing Austenitic Stainless Steel.* Surface Coatings and Technology, vol. 279, pp. 180-185, 2015.
34. **W. Niu, R.S. Lillard, Z. Li, F. Ernst:** *The Mechanism of Passivity in Austenitic Stainless Steel AISI 316L Carburized to Colossal Super Saturation.* Electrochimica Acta (176). 2015 410-419
35. **D. Wang, H. Kahn, F. Ernst, A.H. Heuer:** *NiAl Precipitation in Delta Ferrite Grains of 17-7 Precipitation-hardening Stainless Steel During Low-Temperature Interstitial Hardening.* Scripta Materialia, vol. 108, pp. 136-138, 2015.
36. **D. Wang, C.W. Chen, J.C. Dalton, F. Yang, R. Sharghi-Moshtaghin, H. Kahn, F. Ernst, R.E.A. Williams, D.W. McComb and A.H. Heuer:** *“Colossal” Interstitial Supersaturation of Delta Ferrite in Stainless Steels—I. Low-Temperature Carburization.* Acta Materialia, vol. 86, pp. 193-207, 2015.
37. **X. Zhan and F. Ernst:** *Crystallization of Ni-P Fabricated by Electroless Deposition: Microscopic Mechanism.* Materials Research Society Symposia Proceedings, vol. 1757, pp. mrsf14-1757-uu03-07, 2015.
38. **D. Wang, F. Ernst, H. Kahn, A.H. Heuer:** *Cellular Precipitation at a 17-7 PH Stainless Steel Interphase Interface during Low-Temperature Nitridation.* Metallurgical and Materials Transactions A (45). 2014 1-8

39. X. Gu, G.M. Michal, F. Ernst, H. Kahn, A.H. Heuer: *Numerical Simulations of Carbon and Nitrogen Composition-Depth Profiles in Nitrocarburized Austenitic Stainless Steels*. Metallurgical and Materials Transactions, vol. 45A, pp. 4268-4279, 2014.
40. **X. Gu**, G.M. Michal, F. Ernst, H. Kahn, A.H. Heuer: *Concentration Dependent Carbon Diffusivity in Austenite*. Metallurgical and Materials Transactions, vol. 45A, pp. 3790-3799, 2014.
41. Y. Ge, F. Ernst, H. Kahn, A.H. Heuer: *The Effect of Surface Finish on Low-Temperature Acetylene-Based Carburization of 316L Austenitic Stainless Steel*. Metallurgical and Materials Transactions A, vol. 45, pp. 2338-2345, 2014.
42. D. Wu, H. Kahn, G.M. Michal, F. Ernst and A.H. Heuer: *Orientation Dependence of Nitrogen Supersaturation in Austenitic Stainless Steel During Low-Temperature Gas-Phase Nitriding*. Acta Materialia, vol. 2014, pp. 339-350, 79.
43. **Z. Li**, F. Ernst, A.H. Heuer: *Improved Accident Tolerance of Austenitic Stainless Steel Cladding through Colossal Supersaturation with Interstitial Solutes*. Transactions of the American Nuclear Society, vol. 109, pp. 304-306, 2013.
44. **D. Wang**, H. Kahn, J. J. Lewandowski, F. Ernst, G. M. Michal and A. H. Heuer: *Sustained-load Crack Growth of Hydrogen-Charged Surface-Hardened 316 L Stainless Steel*. Materials Science and Engineering A, vol. 556, pp. 43-50, 2012.
45. **J-P. Hsu**, **D. Wang**, H. Kahn, F. Ernst, G.M. Michal, A.H. Heuer: *Fatigue Crack Growth of Interstitially Hardened 316L Stainless Steel in Air and 0.6 M NaCl*. International Journal of Fatigue, vol. 47, pp. 100-105, 2013.
46. Q. He, B. Shyam, **M. Nishijima**, X. Yang, B. Koel, F. Ernst, D. Ramaker, S. Mukerjee: *Highly Stable PtAuRu/C Catalyst Nanoparticles for Methanol Electro-Oxidation*. Journal of Materials Chemistry, vol. 117, pp. 1457-1467, 2013.
47. **N. Matic**, G. S. Chottiner, F. Ernst, D. Scherson: *The Platinum/Titanium-Nitride Interface: X-Ray Photoelectron Spectroscopy Studies*. Electrochemical and Solid State Letters, vol. 15, pp. 79-82, 2012.
48. M.-C. Hsieh, **Y. Ge**, G.M. Michal, F. Ernst, A.H. Heuer: *Volatility Diagrams for Cr-O and Cr-Cl and Their Application to Removing the Passive Film from the Surface of Stainless Steel*. Metallurgical and Materials Transactions, vol. 43, pp. 1187-1201, 2012.

49. **N. Matic**, G. S. Chottiner, F. Ernst, D. Scherson: *The Platinum/Titanium-Nitride Interface: X-Ray Photoelectron Spectroscopy Studies*. Transactions of the Electrochemical Society, vol. 41, pp. 865-874, 2011.
50. H. Kahn, A.H. Heuer, G.M. Michal, F. Ernst, **R. Sharghi-Moshtaghin**, **Y. Ge**, P.M. Natishan, R.J. Rayne, F.J. Martin: *Interstitial Hardening of Duplex 2205 Stainless Steel by Low-Temperature Carburization: Enhanced Mechanical and Electrochemical Performance*. Surface Engineering, vol. Surface Engineering, pp. 2012, 28. 213-219
51. A. H. Heuer, H. Kahn, F. Ernst, G. M. Michal, D. Hovis, R. J. Rayne, F. J. Martin, and P. M. Natishan: *Enhanced Corrosion Resistance of Interstitially Hardened Stainless Steel: Implications of a Critical Passive Layer Thickness for Breakdown*. Acta Materialia, vol. 60, pp. 716-725, 2011.
52. **D. Wu**, H. Kahn, G.M. Michal, F. Ernst, A.H. Heuer: *Ferromagnetism in Interstitially Hardened Austenitic Stainless Steel Induced by Low-Temperature Gas-Phase Nitriding*. Scripta Materialia, vol. 65, pp. 1089-1092, 2011.
53. F. Ernst, **D. Li**, A. H. Heuer, H. Kahn and G. M. Michal: *The Carbide M_7C_3 in Low-Temperature-Carburized Austenitic Stainless Steel*. Acta Materialia, vol. 59, pp. 2268-2276, 2011.
54. **L. O'Donnel**, G.M. Michal, F. Ernst, H. Kahn, and A.H. Heuer: *Wear Maps for Low Temperature Carburized 316L Austenitic Stainless Steel Sliding Against Alumina*. Surface Engineering, vol. 26, pp. 284-292, 2010.
55. L. Zhu, M. Teo, P.C. Wong, K.C. Wong, **I. Narita**, F. Ernst, K.A.R. Mitchell, S.A. Campbell: *Synthesis and Characterization of a $CoSe_2$ Catalyst for the Oxygen Reduction Reaction*. Applied Catalysis A, vol. 386, pp. 157-165, 2010.
56. A. Heuer, H. Kahn, F. Ernst, G. Michal, R. Rayne, F. Martin, P.M. Natishan: *Carbon-Enhanced Passivity of PH13-8 Mo: A Precipitation-Hardened Martensitic Stainless Steel*. Electrochemical and Solid-State Letters C, vol. 13, pp. 37-39, 2010.
57. **R. Sharghi-Moshtaghin**, Y. Ge, H. Kahn, X. Gu, F. Martin, P. Natishan, R.J. Rayne, G.M. Michal, F. Ernst, and A. H. Heuer: *Low-Temperature Carburization of the Ni-base Superalloy IN718: Improvements in Surface Hardness and Crevice Corrosion Resistance*. Metallurgical and Materials Transactions, vol. 41, pp. 2022-2032, 2010.
58. **K. Du**, F. Ernst, **M. Pelsozy**, J. Barthel, K. Tillmann: *Expansion of Interatomic Distances at the Surface of Platinum Catalyst Nanoparticles*. Acta Materialia, vol. 58, pp. 836-845, 2010.

59. **M. Guinel**, A. Bonakdarpour, B. Wang, P. K. Babu, F. Ernst, N. Ramaswamy, S. Mukerjee, A. Wieckowski: *Carbon-Supported, Selenium-Modified Ruthenium-Molybdenum Catalysts for Oxygen Reduction in Acidic Media*. ChemSusChem, vol. 2, pp. 658-664, 2009.
60. H. Kahn, G. M. Michal, F. Ernst, and A. H. Heuer: *Poisson Effects on X-Ray Diffraction Patterns in Low-Temperature-Carburized Austenitic Stainless Steel*. Metallurgical and Materials Transactions, vol. 40, pp. 1799-1804, 2009.
61. F. Martin, P. Natishan, E. Lemieux, T. Newbauer, R. Rayne, R. Bayles, H. Kahn, G.M. Michal, F. Ernst, and A.H. Heuer: *Enhanced Corrosion Resistance of Stainless Steel Carburized at Low Temperature*. Metallurgical and Materials Transactions, vol. 40, pp. 1805-1809, 2009.
62. G.M. Michal, X. Gu, W.D. Jennings, H. Kahn, F. Ernst, and A.H. Heuer: *Paraequilibrium Carburization of Duplex and Ferritic Stainless Steels*. Metallurgical and Materials Transactions, vol. 40, pp. 1781-1790, 2009.
63. F. Ernst, **A. Avishai**, H. Kahn, **X. Gu**, G. M. Michal, A. H. Heuer: *Enhanced Carbon Diffusion in Austenitic Stainless Steel Carburized at Low Temperature*. Metallurgical and Materials Transactions A, vol. 40, pp. 1768-1780, 2009.
64. J. M. Ziegelbauer, D. Gatewood, A. F. Gullá, M. Guinel, F. Ernst, D. E. Ramaker, S. Mukerjee: *Fundamental Investigation of Oxygen Reduction Reaction on Rhodium Sulfide-based Chalcogenides*. Journal of Physical Chemistry C, vol. 113, pp. 6955-6968, 2009.
65. **K. Du**, F. Ernst, M. Garrels, J. Payer: *Formation of Nickel Nanoparticles in Nickel-Ceramic Anodes of Solid-Oxide Fuel Cells*. International Journal of Materials Research, vol. 99, pp. 548-552, 2008.
66. N. Agarwal, H. Kahn, **A. Avishai**, G. Michal, F. Ernst, and A.H. Heuer: *Enhanced Fatigue Resistance in 316L Austenitic Stainless Steel Due to Low-Temperature Paraequilibrium Carburization*. Acta Materialia, vol. 55, pp. 5572-580, 2007.
67. F. J. Martin, E. J. Lemieux, T. M. Newbauer, R. A. Bayles, P. M. Natishan, H. Kahn, G. M. Michal, F. Ernst, A. H. Heuer: *Carburization-Induced Passivity of 316L Austenitic Stainless Steel*. Electrochemical and Solid State Letters, vol. 10, pp. C76-C78, 2007.
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Invited Presentations

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3. *Alloy Surface Engineering by Infusion of Interstitial Atoms*. Colloquium, Rice University. November 10, 2022.
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6. *Materials Design by Surface Engineering*. Erich Bloch Symposium, University of Buffalo. June 1, 2017.
7. *Surface Engineering by Concentrated Interstitial Solute*. Colloquium, University of Maryland. May 5, 2017.
8. *Basic and Exploratory Research in the Department of Materials Science and Engineering at CWRU*. Army Research Laboratory, Aberdeen, MD, USA. February 14, 2017.
9. *The Department of Materials Science and Engineering*. Colloquium, Chemistry Department, College of Arts and Sciences, CWRU. October 28, 2016.
10. *Metals and Metalworking: DMSE as R&D Partner*. German Metal Working Delegation Meeting, Cleveland, OH, USA. July 12, 2016.
11. *Interdisciplinary Research*. Atotech Cross-Functional Meeting, Neu-Hardenberg, Germany. April 13, 2016.

12. *The Need to See Small: Structure-Property Relationships*. Atotech Cross-Functional Meeting, Neu-Hardenberg, Germany. April 12, 2016.
13. *CWRU and SCSAM*. Atotech Review Meeting, Berlin, Germany. September 28, 2015.
14. *R&D Interests at CWRU*. Atotech Review Meeting, Berlin, Germany. September 28, 2015.
15. *Computational Modeling of Concentration-Dependent Interstitial Diffusion*. Third Symposium on Surface Hardening of Corrosion-Resistant Alloys: Modeling and Structure, held at MS&T-2014, Pittsburg, PA. October 15, 2014.
16. *Low-Temperature Infusion of Interstitial Solutes: A New Concept for Surface Hardening of Steel*. International Symposium on New Trends in Steel Making and Steel Design, Institut für Neue Materialien, Saarbrücken, Germany. November 11, 2013.
17. *Alloy Surface Engineering by Gas-Phase Infusion of Interstitial Solutes*. Colloquium at the University of Birmingham, Birmingham, UK. August 02, 2013.
18. *Alloy Surface Engineering by Colossal Supersaturation with Interstitial Solutes*. Colloquium at the Christian-Albrechts-University, Department of Materials Science, Kiel, Germany. December 17, 2012.
19. *Surface Engineering of Alloys by Colossal Supersaturation with Interstitials*. Colloquium at the Institute of Engineering Innovation School of Engineering, The University of Tokyo, Tokyo, Japan. February 24, 2012.
20. *Colossal Supersaturation with Interstitial Solutes: Physical Principles and Potential for Alloy Surface Engineering*. WPI-AIMR Annual Workshop, Tohoku University, Sendai, Japan. February 2 to 23, 2012.
21. *Colossal Supersaturation: A Powerful Concept for Surface Engineering of Alloys*. ASM Young Member Night, University Club, Pittsburg, USA. February 16, 2012.
22. *Alloy Surface Hardening by Inward Diffusion of Interstitial Solutes*. Spring Seminar Series, Case School of Engineering, Cleveland, Ohio. March 25, 2011.
23. *Colossal Supersaturation with Interstitial Solutes: Perspectives for Surface Engineering of Medical Devices*. Materials Science & Technology 2010, Houston, Texas, USA. October 17 to 21, 2010.
24. *Thermodynamics of Low-Temperature Surface Carburizing*. 18th Congress of the IFHTSE, Rio de Janeiro, Brazil. July 26-30, 2010.

25. *Alloy Surface Engineering by Inward Diffusion of Interstitial Solutes*. Kyoto University, Kyoto, Japan. June 27 to July 5, 2010.
26. *Carbide-Metal Interfaces in Low-Temperature-Carburized Structural Alloys*. XIII International Conference on Intergranular and Interface Boundaries (iib2010), Mie, Japan. June 27 to July 2, 2010.
27. *Strain State of Pt Catalyst Nanoparticles*. The 2nd International Symposium on Advanced Microscopy and Theoretical Calculations (AMTC2), Nagoya, Japan. June 24-26, 2010.
28. *Alloy Surface Engineering by Inward Diffusion of Interstitial Solutes*. The Ohio State University, Columbus, Ohio. May 10, 2010.
29. *Applications Transmission Electron Microscopy in Materials Research*. CUNY - The City University of New York, New York, USA. April 29, 2010.
30. *Perspektiven höchstauflösender Mikrocharakterisierung in der Festkörper- und Materialforschung (Perspectives of High-Resolution Microcharacterization in Solid-State and Materials Research..* Institut für Festkörperforschung, Forschungszentrum Jülich, Germany. November 23, 2009.
31. *Surface Engineering of Medical Devices by Low-Temperature Carburization*. ASM Conference on Opportunities for Next Generation Implants. Cleveland. Ohio USA August 5-7, 2008
32. *Impact of Low-Temperature Carburization on the Microstructure of Structural Alloys*. 35th International Conference On Metallurgical Coatings And Thin Films (ICMCTF) 2008. San Diego, USA. April 28 to May 2, 2008
33. *Colossal Supersaturation with Interstitial Atoms: A New Concept for Surface Engineering of Structural Alloys*. Colloquium at the University of Toledo. Toledo, USA. March 20, 2008
34. *Impact of Applied Electric Fields on Diffusion Reactions at Metal-Oxide Interfaces*. Materials Science & Technology 2006 Conference and Exhibition, Cincinnati, Ohio, USA. October 15-19, 2006.
35. *The Carbide-Austenite Interface in Austenitic Stainless Steels with a Colossal Supersaturation of Carbon*. Materials Science & Technology 2006 Conference and Exhibition, Cincinnati, Ohio, USA. October 15-19, 2006.
36. *Diffusion Reactions at Metal-Oxide Interfaces: The Effect of an Applied Electric Field*. 2006 TMS Annual Meeting & Exhibition, San Antonio, Texas, USA. March 12-16, 2006.
37. *The Microscopy and Microanalysis Facility at Case: Enabling Tools for Nanotechnology*. Functional Nanomaterials. Cleveland, Ohio. USA November 17, 2004

38. *TEM Studies of Cu₂O-Si and Cu₂O-InP Interfaces Made by Epitaxial Electrodeposition*. Microscopy & Microanalysis, Savannah, Georgia, USA. August 1-5, 2004.
39. *Nanoscale Characterization of Epitaxial Cu₂O Films Made by Electrodeposition*. Electronic Materials Conference (TMS), Notre Dame. June 23-25, 2004.
40. *Colossal Supersaturation: A New Concept for Surface Engineering*. The International Conference On Metallurgical Coatings And Thin Films ICMCTF, San Diego, California, USA. April 19-23, 2004.
41. *Colossal Supersaturation with Interstitial Atoms: A New Concept for Advanced Structural Alloys*. Colloquium at the Max-Planck-Institut für Eisenforschung, Düsseldorf, Germany. March 23, 2004.
42. *Microstructure and Phases of Cu-In-Se*. National CIS R & D Team Meeting, Orlando, Florida Solar Energy Center, Cocoa. November 13-14, 2003.
43. *Surface Hardening of Austenitic Stainless Steels by Colossal Supersaturation with Interstitial Solutes*. Structure-Property Relationships in High-Performance Ferrous-Based Systems Possessing Composite-Like Structures, Symposium at Materials Science & Technology 2003, Chicago. November 9-1, 2003.
44. *Alloy Surface Hardening by Colossal Supersaturation with Interstitial Solutes*. Colloquium of the Materials Science and Engineering Program, Rochester Institute of Technology. October 28, 2003.
45. *Colossal Supersaturation -A New Concept for Surface Engineering*. Colloquium of the Department of Materials Science and Engineering. The Ohio State University, Columbus. October 22, 2003.
46. *Liquid-Phase Deposition of α -Cu-In-Se Thin Films*. Space Photovoltaic Research and Technology Conference (SPRAT), Ohio Aerospace Institute. September 16-18, 2003.
47. *Characterization of Materials at the Nanometer Scale: Key to Nanotechnology*. NASA Tutorial on Nanotechnology, NASA Glenn Research Center, Cleveland, Ohio. August 23, 2003.
48. *Assessing Mechanical Properties of Interatomic Bonds Across Heterointerfaces*. Frontiers of Electron Microscopy in Materials Science (FEMMS-2003), Claremont, California. October 5-10, 2003.
49. *Structure of Cu₂O/Si Interfaces, Made by Epitaxial Electrodeposition*. Structure at the Limit of Resolution, Symposium at the 105th Annual Meeting of the American Ceramic Society, Nashville. April 27-30, 2003.

50. *Atomistic Structure and Composition of Grain Boundaries and Heterointerfaces*. Colloquium of the Materials Science and Engineering Department, University of Pittsburgh. March 20, 2003.
51. *Quantitative HRTEM and Modeling of Interfaces*. First Kyoto Workshop on Computational Materials Science, Kyoto (Japan). March 01-04, 2003..
52. *Advances in Assessing the Atomistic Structure of Grain Boundaries and Heterointerfaces*. Colloquium of the Engineering Research Institute, The University of Tokyo, Japan. February 28, 2003.
53. *Atomic Resolution and Bonding of Interfaces*. Colloquium of the Materials Engineering Department, University of Colorado in Boulder. January 28, 2003.
54. *High-Resolution Electron Microscopy*. Workshop on Diagnostics, Imaging and Spectroscopy, Cleveland, Ohio (USA). April 26, 2002.
55. *Assessing Interface Structures and Interfacial Atomic Bonds by Quantitative HRTEM*. Microscopy Society of Northern Ohio, Winter Meeting, Cleveland, Ohio (USA). February 13, 2002.
56. *Analysis of Interface Structure by Quantitative HRTEM*. Colloquium of the Department of Physics and Astronomy, Arizona State University, Tempe, Arizona (USA). December 04, 2001.
57. *Bridging the Gap from Atoms to Applications by Advanced Methods of Transmission Electron Microscopy*. United Engineering Foundation Meeting on "Integrating Materials Science into Engineering Structures and Devices," Lake Arrowhead, California (USA). November 11-15, 2001.
58. *Quantitative Analysis of Interface Structure by HRTEM*. Annual Meeting of the Microscopy Society of America, Long Beach, California (USA). August 05-09, 2001.
59. *Structure and Bonding of Metal/Oxide Interfaces*. Gordon Conference on High-Temperature Corrosion, New London, New Hampshire (USA). July 22-27, 2001.
60. *Atomistic Structure of Interfaces: Progress in Experimental Methods and Modeling*. Case School of Engineering, Faculty Seminar Series 2000-2001, CWRU, Cleveland (USA). May 1, 2001.
61. *Ge-Si Nanostructures for Quantum Effect Electronic Devices*. Microscopy of Semiconducting Materials, Oxford, UK. March 25-29, 2001.
62. *Progress in Materials Science by Synergy of High-Resolution, Analytical, and In-Situ TEM*. Workshop on Future Perspectives of Transmission Electron Microscopy, Stuttgart, Germany. January 17-18, 2001

63. *Self-Assembly of Ge-Si Dots by Coherency Strains*. DSRC/DARPA Workshop on Self-Assembly in Manufacturing, Strategic Analysis Conference Center, Arlington, Virginia. December 6-7, 2000.
64. *Progress in Determination of Defect Structures by Iterative Digital HRTEM Image Matching*. EUREM 12, Brno (Czech Republic). July 9-14, 2000.
65. *Grain Boundaries in Strontium Titanate*. Japan Fine Ceramics Center Workshop on Grain Boundaries in Ceramics. Nagoya (Japan), March 16-17, 2000.
66. *Structure, Composition, and Bonding of Grain Boundaries in Metals*. 37th Metallurgists' Day and 53rd Annual Technical Meeting of the Indian Institute of Metals. Kanpur (India) November 14-16, 1999.
67. *Atomistic Structure of Interfaces: Progress in Experimental Techniques and Modeling*. 3rd Slovenian-German Seminar, Bled, Slovenia. November 19-21, 1998.
68. *Ge-Si-Nanostructures for Quantum Effect Electronic Devices*. Technion-Israel Institute of Technology, Haifa, Israel. June 14, 1998.
69. *High Precision Assessment of Interface Structures by HRTEM*. 32nd Annual Meeting of the Israel Society for Microscopy, The Weizmann Institute of Science, Rehovot, Israel. June 11, 1998.
70. *Analysis of Interface Structures by Quantitative HRTEM*. MCEM 97, Portoroz, Slovenia. October 5-8, 1997.
71. *Analysis of Interface Structures by Quantitative HRTEM*. EMAG 97, Cambridge, Great Britain. September 2-5, 1997.
72. *Solving Interface Structures by Quantitative High-Resolution Electron Microscopy*. Physical Colloquium, University of Vienna, Austria. December 11, 1997.
73. *Analysis of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Fall Meeting of the Materials Research Society, Boston. December 1-5, 1996.
74. *Quantitative HRTEM of Interfaces and Structure of Metal/Ceramic Interfaces*. Micromat 1996. Rio de Janeiro, October 13-16, 1996.
75. *Analysis of Grain Boundary Structures by Quantitative HRTEM*. Annual Meeting of the Microscopy Society of America (MSA), Minneapolis. August 11-15, 1996.
76. *Interface Dislocations*. International Workshop on Plastic Deformation of Structural Materials at High Temperatures, Schloss Ringberg, Tegernsee, Germany. July 14-17, 1996.

77. *Quantitative High-Resolution Electron Microscopy of Interfaces*. 7th International Conference on Intergranular and Interphase Boundaries in Materials (iib95), Lissabon, Portugal. June 26-29, 1995.
78. *Solving Interface Structures by High-Resolution Electron Microscopy and Computer Modeling*. Colloquium of the Institut für Physikalische Chemie und Elektrochemie, University of Hannover, Germany. May 8, 1995.
79. *Solving Interface Structures by High-Resolution Electron Microscopy and Computer Simulation*. 13th International Congress on Electron Microscopy, Paris, France. July 18-22, 1994.
80. *Quantitative HRTEM of Twin Boundaries*. Annual Congress of the German Physical Society, Münster, Germany. March 21-25, 1994.
81. *Quantitative High-Resolution Electron Microscopy as a Method to Determine the Structure of Grain Boundaries*. German Physical Society, Spring Meeting, Münster. March 21-25, 1994.
82. *Lattice Mismatch Accommodation by Misfit Partial Dislocations*. Annual Meeting of The Materials Society (TMS), San Francisco. February 27-March 3, 1994.
83. *Quantitative HRTEM of Grain Boundaries*. International Symposium on High-Voltage and High-Resolution Electron Microscopy, Stuttgart, Germany. February 21-24, 1994.
84. *Assessment of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Workshop on Microcharacterization of Materials and Devices, Göttingen, Germany. January 6-7, 1994.
85. *Quantitative Evaluation of HRTEM Images*. Multinational Congress of the Austrian, German, and Swiss Electron Microscopy Societies, Zürich. Switzerland, September 1993..
86. *HREM of Metal/Ceramic Interfaces*. Annual Meeting of the French Society of Electron Microscopy (SFME), Lyon, France. June 29-July 4, 1993..
87. *Misfit Dislocation Networks in 111-GeSi/Si-Interfaces*. Colloquium of the IV. Physikalisches Institut, University of Göttingen. June 24, 1993.
88. *Relaxation of Misfit Stresses in Epitaxial Layers*. Physical-Chemical Colloquium, University of Stuttgart, Germany. June 22, 1993.
89. *Phase Transformations at Metal/Ceramics Interfaces*. International Workshop on Reactive Phase Formation at Interfaces and Diffusion Processes, Aussois, France. May 21-2, 1993..

90. *Misfit Dislocation Structure of GeSi/Si-{111}-Interfaces*. European Science Foundation Conference on Extended Defects in Semiconductors, Holzgau, Germany. August 29-September 3, 1992.
91. *Quantitative Analysis of Intensities in HRTEM Lattice Images*. Annual Congress of the German Electron Microscopy Society (DGE), Darmstadt, Germany. September 8-13, 1991.
92. *Misfit Dislocation Networks in GeSi/Si Interfaces Grown by Liquid Phase Epitaxy*. Colloquium of the Department of Materials Science and Engineering, CWRU, Cleveland, USA. February 11, 1991.
93. *HRTEM of the Cu/MnO Interface in an Internally Oxidized CuMn Alloy*. Colloquium of the Department of Materials Science and Engineering, CWRU, Cleveland, USA. April 23, 1990.
94. *HRTEM of Metal/Ceramic Interfaces in Internally Oxidized Alloys*. 4th Workshop on Characterization and Properties of Interfaces in Crystals, Delft, Netherlands. June 5-9, 1989.
95. *High-Resolution Electron Microscopy of Semiconductor Heterointerfaces*. Colloquium of the Max-Planck-Institut für Festkörperforschung, Stuttgart. Germany. May 8, 1989.
96. *High-Resolution Electron Microscopy of Cu/Al₂O₃ Interphase Interfaces*. Colloquium of the Institut für Metallphysik, University of Göttingen, Germany. April 18, 1989.

Courses Taught and Developed*

Univ	Code	Course
Stuttgart		Kristallografie und Beugung* (Crystallography and Diffraction)
Stuttgart		Werkstoff-Analytik 1* (Analytical Methods of Materials Science 1*)
Stuttgart		Halbleiter Werkstoffe der Mikorelektronik und Photovoltaik* (Semiconductor Materials for Microelectronics and Photovoltaics*)
CWRU	EMSE-102	Introduction to Materials Science and Engineering*
CWRU	EMSE-228	Mathematical and Computational Methods for Materials Science and Engineering*
CWRU	EMSE-290	Materials Laboratory III
CWRU	EMSE-314	Electrical, Optical, and Magnetic Properties of Materials*
CWRU	EMSE-327	Thermodynamic Stability and Rate Processes*
CWRU	EMSE-368/468	Scientific Writing*
CWRU	EMSE-398-399	Senior Project in Materials Science and Engineering
CWRU	EMSE-465	Surface Engineering of Materials*
CWRU	EMSE-509	Conventional Transmission Electron Microscopy*
CWRU	EMSE-512	Advanced Methods of Transmission Electron Microscopy*
CWRU	EMSE-515	Analytical Methods of Materials Science and Engineering*

Consulting

Period	Client	Activity
Since 07/2026	Cooley LLP	Expert Witness for Tower in: Global-Foundries vs Tower Semiconductor
08/2025- 01/2026	Cooley LLP	Expert Witness for Caihong in: Corning Incorporated vs Caihong in vs Display Devices Co., Ltd., d/b/a Irico Display Devices Co., Ltd.
03/2025- 07/2025	Merchant & Gould P. C.	Expert for Caihong in: Inter Partes Review Petitions of Caihong in vs Display Devices Co., Ltd., d/b/a Irico Display Devices Co., Ltd. vs Corning Incorporated
Since 02/2026	Bosch	Consultant for Semiconductor Device Processing
02/2025- 12/2025	Bosch	Consultant for Semiconductor Device Processing
07/2018- 01/2024	Reichman & Jorgensen	Expert Witness for Arconic in: Arconic Corporation et al vs Novelis Inc. et al
05/2018- 08/2018	Voss Aerospace In- dustries Inc.	Mechanical Properties of Aluminum Alloys
01/2017- 12/2024	KFAS	Proposal Reviewer for Kuwait Foundation for the Advancement of Sciences
03/2016- 04/2018	Atotech	Development of a Know-How Management System
03/2014- 08/2021	Skolkovo	Proposal ("Expert") Reviewer
04/2014- 06/2014	Voss Aerospace In- dustries Inc.	Microstructure of Precipitation-Hardened Aluminum Alloys

07/2012	Powdermet	Microscopic Structure of Novel Powders
07/2011– 06/2012	Atotech	Microstructure of Electroplated Cu for Microelectronic Devices
04/2011– 10/2011	AlSher Titania LLC	Structure and Composition of Titania Nanoparticles
01/2011– 12/2017	Pocared	Microstructure of Coatings for Pharmaceutical Containers
09/2008	TRW Automotive Engine Components	Nanohardness and Auger Electron Spectroscopy Studies of Valves
04/2008	DoD	Proposal reviewer for SERDP program
04/2007	DoD	Proposal reviewer for SERDP program
10/2006– 05/2007	Bird Technologies Group	Scanning Electron Microscopy and Analysis of Resistors
11/2004– 04/2005	Patent Litigation	Expert Witness Assistance for Experimental Studies of Coatings
05/2004– 11/2004	OMG Americas	Transmission Electron Microscopy Analysis of Novel Nanoparticles
12/2003– 06/2005	Moen Incorporated, North Olmsted, Ohio	Transmission Electron Microscopy Studies of Coatings
06/2003– 12/2003	IMQ Corporation, Chesterland, Ohio	Transmission Electron Microscopy Studies on the Effect of Boron on the Surface Structure of Ferritic Steels.
02/2003– 05/2003	Inst. of Semiconductor Technology, Stuttgart, Germany	Transmission Electron Microscopy Analysis of Defect Structures in Ultra-thin SiGe Buffer Layers